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SANTA CLARA UNIVERSITY	ELEN 115 Spring 2024	Shoba Krishnan
Project # 1: Sensors		

I. OBJECTIVES

- To learn how to configure a sensor.
- To learn how to read a data sheet to understand device behavior
- To design an experiment to study the characteristics of a sensor

II. Pre-lab

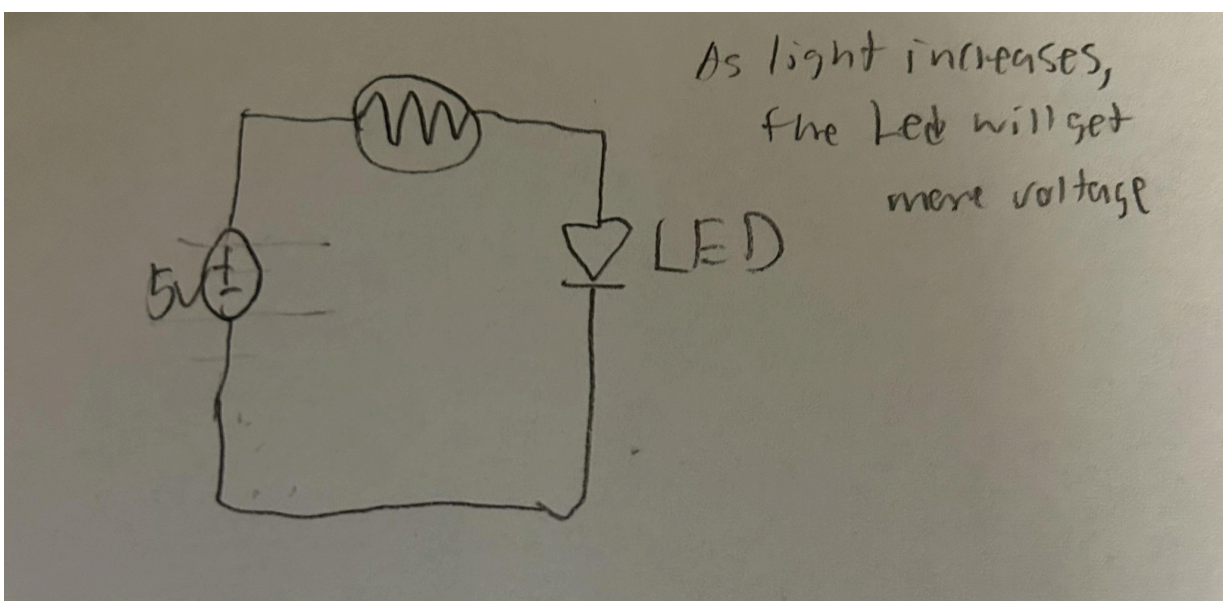
Part 1: Sensor 1

To obtain a electrical voltage or current to feed into an electronic circuit we need to sense physical quantities. There are many type of sensors, but in this lab we use a photoresistor.

Figure 1: Photoresistor

Explain the behavior of a photoresistor. Draw a schematic of a simple circuit that uses the photoresistor and can help detect the presence of varying amounts of light.

A photoresistor lowers its resistance as more light shines on it. The light excites bound electrons which is how resistance goes down.



Part 2: passive filters

Noise interference on photoresistor

All sensor inputs sometimes have noise from the environment.

The sensor output is a DC voltage generated from the photoresistor configured as in Part I. There is a noise signal in the KHz range that is interfering with the sensor data. What type of filter would you need that provides only the sensor output signal that is DC after removing this noise.

A low pass filter. **The dc signal has frequency 0 and thus would be allowed through but the KHz (high frequency) signal would get blocked.**

If the sensory output is a result of pulsing light at a frequency in tens of Khz, but there is some low frequency ambient noise, what type of filter would you need that provides only the sensor output signal after removing this noise.

A high pass filter. This will allow the high frequency pulse through but not the low frequency noise.

Part 3: Sensor 2

> We would like to study a flex sensor for the lab. Look up the data sheet of a flex sensor and explain how it works. Give a possible application for such a sensor.
>
> **A flex sensor measures how much it's bending by increasing resistance as it gets more bent. Applications include hinge sensors and robotics where you would want to know how much something bends.**
>
> In lab you will be designing an experiment to characterize a flex sensor. Come prepared with some ideas.
>
> **Ok** 👍